

DESTINATION INDIA FOR ELECTRONICS DESIGN

The availability of skilled talent and cost advantages are driving the growth of chip design services in India

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If China is considered the production house of the world, India can draw a bead on becoming the design capital. Say hello to a small but growing phenomenon that is steadily finding its place in the sun—electronics design service. Indian companies are leaders here.

A tectonic shift is underway in the high-technology world. Chip manufacturing activities moved out of US markets around 15 years ago in favour of Asian foundries, and now the back-end, verification and design services have also begun their eastward march. The centre of gravity of the high-tech industry is steadily moving eastwards, opening the doorway to huge opportunities for Asian nations, particularly India.

According to the global research firm Research and Markets, India's design service industry will continue to grow at double-digit growth until

2010. Communications and consumer electronics are the major growth drivers of the design services market in India. The cost advantage, availability of skilled manpower and software process maturity have compelled original design manufacturers (ODMs) to either outsource part of their design activities to third-party design firms in India or set up their own captive centres in the country. The captive centres of the ODMs work on cutting-edge technologies and the latest process nodes and contribute to the development of products for a global customer base.

However, the industry is facing a lot of challenges, like increasing salaries and lack of analogue/mixed-signal expertise in the country. The companies are building strengths to overcome these challenges and explore new opportunities.

How the market looks like?

Currently, India is emerging as a key destination for integrated circuits

(ICs), field-programmable gate arrays (FPGAs) and system-on-a-chip (SoC) designs. According to Indian Semiconductor Association (ISA), a rough estimate of the types of designs being done in India is ASICs 40 per cent, SoCs 20 per cent, structured ASICs 5 per cent, gate array 10 per cent, CPLDs/FPGAs 15 per cent and fully customised products 10 per cent.

The market is huge. The 'fables' trend continues to soar and the US Fabless Semiconductor Association predicts that the market shares of fables companies will jump from just over 10 per cent today to 50 per cent by 2010. Additionally, there is the IP design market opportunity.

The Indian design services industry will maintain an impressive growth rate of 20.2 per cent during the 2007–12 timeframe as against a 6 per cent year-on-year growth globally, and reach US\$3.4 billion by 2012.

The electronics design ecosystem consists of many elements including captive centres of ODMs like Cadence, Synopsys, Mentor Graphic, Magma and Texas Instruments; subsidiaries of Indian OEMs like NXP, TATA Elxsi and Freescale Semiconductor; independent engineering design firms like Rabyte, Evolute, Alpha Design, SLN Technologies and Jupiter Electronics; and Indian design services firms like TCS, Wipro, Infosys, Tata Elxsi and KPIT Cumins.

The VLSI design activity in India is concentrated in mainly four cities: Bengaluru followed by Hyderabad, Delhi/Noida and Chennai. Bengaluru is now home to 70 of the 130 firms (including multinationals) engaged in chip design in India, making it comparable to

